

LINMA2222 | Project

Part 4: MSPE

Instructions

This project is due on **December 12, 2025, at 11:59 pm**. You should keep the same teams as in Part 1. As a reminder, each team should upload a single zip-file, containing their report and code, on Moodle. Make sure to clearly state the team members in your report. You can do the project in MATLAB *or* Python *or* Julia, but not a mix.

For your convenience, we have printed Parts 1, 2 and 3 below in gray. Part 4 starts at Section 8.

1 Introduction

This project studies a *portfolio optimization problem*. Given a financial asset a , with price p_t at time t , the goal is to determine an optimal strategy for buying or selling units of a , based on the current portfolio state and available information, at each discrete time step $t = 0, 1, 2, \dots$.

2 The model

Let $q_t \in \mathbb{R}$ denote our *inventory* at time t , that is, the quantity of asset a (e.g., a stock, gold, etc.) we hold at time t , expressed as a fraction of the total market supply (e.g., 1000 stocks/units). For simplicity, we assume that the decision of buying or selling inventory is taken only at discrete time intervals (e.g., at the beginning or end of each day); hence, we let $t \in \mathbb{N}$.

Let $p_t \in \mathbb{R}$ be the *market price* at time t (this is the price that we observe and we use to make decisions), and let $\bar{p}_{t,t+1} \in \mathbb{R}$ be the average execution price on the interval $[t, t+1)$ (this is the price at which the trade actually takes place, which may differ from p_t).

We consider the following *gross* stage reward, which measures our profit made from trading:

$$g_t := 1000 \cdot (q_{t+1}p_{t+1} - q_t p_t - (q_{t+1} - q_t)\bar{p}_{t,t+1}). \quad (1)$$

Question 2.1 Provide an interpretation of g_t . What does it represent financially?

Given a *utility function* $c : \mathbb{R} \rightarrow \mathbb{R}$, we consider the following stage reward (sometimes called *net* or *utile* stage reward):

$$r_t = c(g_t). \quad (2)$$

The utility function allows to capture different phenomena, like taxes, aversion to risk, etc.

The variable we can control, hereafter called *control input*, is denoted by $u_t \in \mathbb{R}$ and represents the quantity of the asset a (again as a fraction of the asset's total availability) that we buy ($u_t \geq 0$) or sell ($u_t \leq 0$) at time $t \in \mathbb{N}$. Hence, for all $t \in \mathbb{N}$,

$$q_{t+1} = q_t + u_t.$$

Objective The goal of a trader is to find policies that maximize the infinite-horizon average reward:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=0}^{T-1} r_t \right]. \quad (3)$$

The admissible policies take the form $u_t = \pi(x_t)$ (if deterministic) or $u_t \sim \pi(\cdot | x_t)$ (if randomized), where x_t is the vector of system state variables, described in the model below.

$\theta = 0.5$	$\omega^a = 0.1$	$\omega^u = 0.2$
$\sigma^a = 0.018$	$\beta^u = -0.048$	$\gamma^u = 0.06$
	$\sigma^p = 0.02$	

Table 1: Model parameters.

Model We next describe the model that governs the evolution of the system. The state vector is $x_t = [q_t, z_t^a, z_t^u]^\top$, where q_t is the inventory as above, and z_t^a and z_t^u are two other variables modeling components (one stochastic and one input-dependent) of the variation of the price p_t , and described next:

- The evolution of z_t^a is given by

$$z_{t+1}^a = (1 - \omega^a)z_t^a + \omega^a \sigma^a \xi_t^a, \quad \xi_t^a \sim \mathcal{N}(0, 1),$$

where $\omega^a \in (0, 1)$ and $\sigma^a \in \mathbb{R}_{\geq 0}$ are parameters.

- The evolution of z_t^u is given by

$$z_{t+1}^u = (1 - \omega^u)z_t^u + \omega^u \beta^u u_t,$$

where $\omega^u \in (0, 1)$ and $\beta^u \in \mathbb{R}$ are parameters.

- The evolution of the price p_t is given by

$$p_{t+1} = p_t + z_t^a + z_t^u + \gamma^u u_t + \sigma^p \xi_t^p, \quad \xi_t^p \sim \mathcal{N}(0, 1),$$

where $\gamma^u \in \mathbb{R}$ and $\sigma^p \in \mathbb{R}_{\geq 0}$ are parameters.

- The execution price $\bar{p}_{t,t+1}$ is given by

$$\bar{p}_{t,t+1} = \theta p_t + (1 - \theta)p_{t+1} + \theta \gamma^u u_t,$$

where $\theta \in (0, 1)$ is a parameter.

We stress that, in the above dynamics, $\{\xi_t^a\}_{t \in \mathbb{N}}$ and $\{\xi_t^p\}_{t \in \mathbb{N}}$ are independent random variables with standard normal distribution. We let $\xi_t = [\xi_t^a, \xi_t^p]^\top$ for all $t \in \mathbb{N}$.

Question 2.2 Provide an interpretation of the model and explain why it offers a reasonable representation of the evolution of an asset's price over time.

Question 2.3 Express the gross stage reward g_t in (1) as a function of x_t , u_t and ξ_t . In particular, compute explicitly the matrix H such that $g_t = \frac{1}{2}[x_t^\top, u_t^\top, \xi_t^\top]H[x_t^\top, u_t^\top, \xi_t^\top]^\top$.

Utility function Finally, we focus on the following instance for the utility function:

$$c(g) = \max \left(g - \frac{1}{2}g^2, 1 - \exp(-g) \right). \quad (4)$$

Question 2.4 Show that the utility function above is penalizing gross stage rewards g_t with a large variance. More precisely, suppose g_t is a random variable with some distribution of your choice (e.g., normal, uniform, etc.) with zero mean and variance σ^2 ; show empirically and/or analytically that $\mathbb{E}[c(g_t)]$ is a decreasing function of σ^2 .

3 Closed-loop analysis

In this subsection, we use the parameters given in Table 1.

3.1 Unconstrained case

We consider the policy $\pi_{\text{cl}}(x_t) = K_{\text{cl}}x_t$, where $K_{\text{cl}} = [-0.5, 0.5, 0.5]$.

Question 3.1 Plot a sample trajectory of the closed-loop system with policy π_{cl} , starting at $x_0 = 0$, with horizon $T = 1000$ (i.e., for $t = 0, 1, \dots, T$). In your plots, show the variables x_t and u_t , as well as the average reward $\frac{1}{t} \sum_{s=0}^{t-1} r_s$, for $t = 0, 1, \dots, T$. Interpret the results.

Question 3.2 Generate $N = 100$ sample trajectories, each with horizon $T = 1000$, of the closed-loop system with policy π_{cl} , starting at $x_0 = 0$. For each of them, compute the average reward $\frac{1}{t} \sum_{s=0}^{t-1} r_s$, for $t = 0, 1, \dots, T$. Show in a plot the mean and variance (over the N samples) of the latter for $t = 0, 1, \dots, T$. Interpret the results.

3.2 Constrained case

The policy considered in the previous subsection may be infeasible, since it could lead to an inventory q_t that is smaller than zero (meaning that the investor holds a negative amount of the assets) or larger than one (meaning that the investor holds more than 100% of the market availability). Therefore, we next consider a *projected policy*, given by

$$\pi_{\text{pcl}}(x_t) = \max(-q_t, \min(1 - q_t, \pi_{\text{cl}}(x_t))).$$

Question 3.3 Repeat question 3.1 for π_{pcl} .

Question 3.4 Repeat question 3.2 for π_{pcl} .

3.3 A better policy?

Finally, propose a policy π_{yours} that satisfies the constraints on q_t ($0 \leq q_t \leq 1$) and potentially provides a better infinite-horizon average reward. You may use any reasoning you prefer (your investor intuition, a mathematical model, or otherwise), provided that it is justified.

Question 3.5 Describe and motivate *briefly* the policy π_{yours} . Show that it does better than the proposed policy (if so!).

Question 3.6 Repeat question 3.1 for π_{yours} .

Question 3.7 Repeat question 3.2 for π_{yours} .

4 LQR approximate model

In this section, we derive an LQR *approximate* model of the system, and solve the associated optimal control problem. Notice that some notation and equation numbers refer to Part 1.

4.1 Derivation of the model

First, note that the dynamic of the system is linear. Therefore, it can be expressed in explicit linear form:

$$x_{t+1} = Fx_t + Gu_t + D\xi_t,$$

where F , G and D are matrices of appropriate dimensions and $x_t = [q_t, z_t^a, z_t^u]^\top$.

Question 4.1 Derive the matrices F , G , and D above.

Consider the utility function

$$c_{\text{quad}}(g) = g - \frac{1}{2}g^2. \tag{5}$$

Question 4.2 Derive matrices S, R, P such that

$$\mathbb{E}[c_{\text{quad}}(g_t) | x_t, u_t] = \frac{1}{2}x_t^\top Sx_t + x_t^\top Pu_t + \frac{1}{2}u_t^\top Ru_t + o(x_t^2 + u_t^2).$$

Hint. Use Question 2.3, but do not neglect the effect of the utility function.

Question 4.3 (Bonus) Recall that the stage reward is defined in (2) as $r_t = c(g_t)$ where c is as in (4). Show that $\mathbb{E}[c_{\text{quad}}(g_t) | x_t, u_t]$ is a quadratic approximation of $\mathbb{E}[r_t | x_t, u_t]$ in the sense that

$$\mathbb{E}[c_{\text{quad}}(g_t) | x_t, u_t] = \mathbb{E}[r_t | x_t, u_t] + o(x_t^2 + u_t^2).$$

LQR approximate model Given matrices S , P and R of appropriate dimensions, define the *approximate* reward by

$$\hat{r}(x_t, u_t) = \frac{1}{2}x_t^\top Sx_t + x_t^\top Pu_t + \frac{1}{2}u_t^\top Ru_t. \quad (6)$$

The system with objective to maximize the infinite-horizon average approximate reward,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=0}^{T-1} \hat{r}_t(x_t, u_t) \right], \quad (7)$$

is called the *LQR approximate model* of the system.

4.2 Optimal controller

In this subsection, we use the parameters given in Table 1.

Remark 1. All numerical experiments should be done on the true system, that is, using the true stage reward model $r_t = c(g_t)$ where c is as in (4).

Question 4.4 Using tools from the course, compute the optimal policy for the LQR approximate model. Denote this policy by π_{lqr} . Also, give the value of the infinite-horizon average approximate reward (7) with this policy.

Question 4.5 Repeat question 3.1 for π_{lqr} . Compare (graphically) the latter with the optimal infinite-horizon average approximate reward found in Question 4.4.

Question 4.6 Repeat question 3.2 for π_{lqr} . Compare (graphically) with the optimal infinite-horizon average approximate reward found in Question 4.4.

Since the policy π_{lqr} may violate the constraint that $q_t \in [0, 1]$ for all $t \in \mathbb{N}$, we consider the *projected* policy defined by $\pi_{\text{plqr}}(x_t) = \max(-q_t, \min(1 - q_t, \pi_{\text{lqr}}(x_t)))$.

Question 4.7 Repeat question 3.1 for π_{plqr} .

Question 4.8 Repeat question 3.2 for π_{plqr} .

A less suboptimal way to enforce the constraint $q_t \in [0, 1]$ for all $t \in \mathbb{N}$ consists in using MPC. Consider the MPC optimization with horizon $\mathcal{N}$ and noiseless prediction model:

$$\begin{aligned} \min_{\substack{\hat{u}_0, \dots, \hat{u}_{\mathcal{N}-1} \in \mathbb{R} \\ \hat{x}_1, \dots, \hat{x}_{\mathcal{N}} \in \mathbb{R}^3}} \quad & \hat{x}_{\mathcal{N}}^\top R_0 \hat{x}_{\mathcal{N}} + \sum_{k=0}^{\mathcal{N}-1} \hat{r}_k(\hat{x}_k, \hat{u}_k) \\ \text{s.t.} \quad & \hat{x}_0 = x_t, \\ & \hat{x}_{k+1} = F\hat{x}_k + G\hat{u}_k, \quad k = 0, \dots, \mathcal{N}-1, \\ & 0 \leq [1 \ 0 \ 0] \hat{x}_k \leq 1, \quad k = 0, \dots, \mathcal{N}, \end{aligned}$$

where $\hat{r}$ is as in (6). Let π_{mpc} denote the corresponding policy.

Question 4.9 Justify your choice of $\mathcal{N}$.

Question 4.10 Repeat question 3.1 for π_{mpc} .

Question 4.11 Repeat question 3.2 for π_{mpc} .

5 Exact policy improvement

In this section, we use the parameters given in Table 1.

Question 5.1 Consider the LQR approximate model obtained in Section 4, i.e., with objective (7). Ignore the *process* noise terms (i.e., set $\sigma^a = \sigma^p = 0$), so that the system becomes deterministic. Change the objective in (7) by the infinite-horizon *total* reward, i.e., remove the factor $\frac{1}{T}$ in (7). Implement the Exact Policy Iteration Algorithm (E-PIA) (cf. Exercise 3.1 in the lecture notes). As initial policy, use $K_0 = K_{cl}$, where K_{cl} is defined in Section 3. Plot the evolution of the error norm between K_k and K_{lqr} (where K_{lqr} is the gain associated with the policy π_{lqr}) as a function of k . Check that the sequence K_k converges to K_{lqr} .

6 Least-square temporal difference

We consider an approximate Q-function, denoted Q^θ , parameterized by a parameter vector $\theta \in \mathbb{R}^d$.

Question 6.1 In the lecture notes, Proposition 3.1 discusses the least-square formulation for estimating the parameter vector θ , when Q^θ is linearly parameterized (i.e., $Q^\theta = \theta^\top \psi$). Provide a detailed proof of Proposition 3.1. Note that this is Proposition 5.3 from Meyn's book.

Question 6.2 (Interpretation and basis functions for LSTD) What are some reasonable choices for the basis functions ψ used in the LSTD algorithm? Discuss how these choices could influence the performance of the algorithm. *Hint: think of policy evaluation and policy improvement.* Suppose you know the system is linear and the objective is quadratic, what would you pick for the basis functions ψ ?

6.1 Implementation of LSTD

In this subsection, we use the parameters given in Table 1.

Question 6.3 Implement the LSTD algorithm to estimate the Q-function of the closed-loop system with policy π_{cl} . Ignore the *process* noise terms (i.e. set $\sigma_a = \sigma_p = 0$) so that the system becomes deterministic (and therefore, change the objective in Eq. 3 by the infinite-horizon *total* reward, i.e., remove the factor $\frac{1}{T}$ in Eq. 3). To generate your data, start from a non-zero initial condition (why? comment your choice) and use an exploration policy π_{exp} , e.g., use $\pi_{exp}(\cdot | x) \sim \mathcal{N}(K_{cl}x, \sigma_{exp}^2)$.

Use LSTD to fit a Q-function $Q^\theta = \theta^\top \psi$ using this data, for a reasonable choice of basis functions ψ and exploration noise σ_{exp} (comment on its effect on the solution). Comment and justify your results. In particular, compare your learned Q-function with an estimation $\hat{Q}$ that you can get from simulating a long enough trajectory of the system. Namely, for different values of (x, u) (e.g., obtained with the exploration policy), plot $Q^\theta(x, u)$ against $\hat{Q}(x, u)$.

Question 6.4 Consider the *LQR approximate model* obtained in Section 4, i.e., with objective (7). Ignore the *process* noise terms (i.e., set $\sigma^a = \sigma^p = 0$), so that the system becomes deterministic. Change the objective in (7) by the infinite-horizon *total* reward, i.e., remove the factor $\frac{1}{T}$ in (7). For this model, repeat Question 6.3. Is there another way to obtain true values of $Q(x, u)$ for this particular system?

6.2 Implementation of LSPI

In this subsection, we use the parameters given in Table 1.

We can now combine the learning of the Q-function (based on LSTD) with the policy improvement step, overall known as Least-Squares Policy Improvement (LSPI).

Question 6.5 For the setting of Question 6.3, implement the LSPI algorithm. Show the plot of total reward of the closed-loop system with your obtained policy (say π_{lspi}) against those with π_{cl} and π_{lqr} for different initial conditions (e.g., choose three or four such initial conditions that you find relevant). Comment and justify your results.

Question 6.6 For the setting of Question 6.4, implement the LSPI (or LSTD-PIA) algorithm. Demonstrate the convergence of the gain sequence K_k to K_{lqr} (where K_{lqr} is the gain associated to π_{lqr}) by plotting the evolution of the error norm with k .

Finally, since the policies may violate the constraint $q_t \in [0, 1]$ for all $t \in \mathbb{N}$, we incorporate this constraint in the policy improvement step. Therefore, the policy improvement consists in solving the optimization problem

$$\pi_\theta(x) = \arg \max_u Q^\theta(x, u) \quad \text{s.t.} \quad 0 \leq [1 \ 0 \ 0] x_k + u \leq 1, \quad (8)$$

Question 6.7 Repeat Question 6.5 (i.e., using the true reward model of the system) but with the constrained policy improvement step discussed above.

7 Q-learning

In this section, we use the parameters given in Table 1.

Question 7.1 Consider the *LQR approximate model* obtained in Section 4, i.e., with objective (7). Ignore the *process* noise terms (i.e., set $\sigma^a = \sigma^p = 0$), so that the system becomes deterministic. Change the objective in Eq. 7 by the infinite-horizon *total* reward, i.e., remove the factor $\frac{1}{T}$ in (7). Estimate the sequence of policy gains K_k using the $Q(\lambda)$ -learning algorithm. Discuss the convergence of the policy gains K_k with respect to the optimal LQR gain K_{lqr} . Finally, discuss the effect of the trace-decay parameter $\lambda \in [0, 1]$ on the convergence rate and accuracy of the learned policy.

Question 7.2 Consider the true system. Ignore the *process* noise terms (i.e. set $\sigma_a = \sigma_p = 0$) so that the system becomes deterministic (and therefore, change the objective in Eq. 3 by the infinite-horizon *total* reward, i.e., remove the factor $\frac{1}{T}$ in Eq. 3). Apply the $Q(\lambda)$ -learning algorithm on this system. Show the plot of total reward of the closed-loop system with your obtained policy (say π_Q) against those with π_{lspi} , π_{cl} and π_{lqr} for different initial conditions (e.g., choose three or four such initial conditions that you find relevant). Comment and justify your results.

8 Mean-square Poisson error

You have seen in the course that LSTD, used in Part III of the project, can be formulated as a linear system of equations to be solved in the least-square sense. Consider now the Poisson error (PE) for the Q -function of the closed-loop system with policy π :

$$\mathcal{P}^\theta(x_t, u_t) := \mathbb{E}[r_t + Q^\theta(x_{t+1}, \pi(x_{t+1})) | x_t, u_t] - \eta^\theta - Q^\theta(x_t, u_t),$$

where η^θ is an approximation of the averaged reward and Q^θ is an approximation of the Q -function of the closed-loop system with policy π .

Question 8.1 Compare the Poisson error described above with the Bellman error seen in the course for deterministic systems and for stochastic systems with discounted cost.

The Mean-Square Poisson Error (MSPE) is defined as

$$\mathbb{E}_{(x,u) \sim \mu} [\mathcal{P}^\theta(x, u)^2],$$

where μ is a probability distribution on $\mathcal{X} \times \mathcal{U}$ (e.g., the steady-state measure of the system controlled with the exploration policy).

Question 8.2 (Mean-square Poisson error as a linear system) Show that if Q^{θ_1} and Q^{θ_2} are linearly parametrized (i.e., $Q^{\theta_1}(x, u) = \theta_1^\top \psi_Q(x, u)$ for some $\psi_Q : \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}^d$ and $\eta^{\theta_2} = \theta_2$), then minimizing the MSPE can be formulated as a linear system of equations to be solved in the least-square sense. Compute the matrix A_{lspe} and the vector b_{lspe} of the corresponding normal equations. Discuss the differences with the LSTD framework used in Part III.

Hint: To get the linear system of equations, define $\theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} \in \mathbb{R}^{d+1}$.

Question 8.3 (Estimation of MSPE linear system) Propose an approximation of A and b in Question 8.2 from data obtained from the system controlled by some exploration policy π_{exp} , assuming that μ is the steady-state measure of the system controlled by π_{exp} . The method does not need to be complicated. You may assume that you have a subroutine to compute conditional expectations of the form

$$\mathbb{E}[f(x_{t+1}, \pi(x_{t+1})) \mid x_t = x, u_t = u].$$

Then, explain how such a subroutine can be implemented if you have the transition kernel $\kappa(x' \mid x, u)$ of the system, defined by

$$\mathbb{P}[x_{t+1} \in A \mid x_t = x, u_t = u] = \int_A \kappa(x' \mid x, u) dx,$$

and the policy π .

We call *Least-Squares Poisson Error* (LSPE) the method developed in Questions 8.2 and 8.3.

8.1 Implementation of LSPE

In this subsection, we use the parameters given in Table 1.

You may assume that the transition kernel of the system has *finite support*, i.e., for each $(x, u) \in \mathcal{X} \times \mathcal{U}$, there is $\{x'_1(x, u), \dots, x'_m(x, u)\} \subseteq \mathcal{X}$ and $\{p_1(x, u), \dots, p_m(x, u)\} \subseteq [0, 1]$ such that

$$\kappa(x' \mid x, u) = \sum_{i=1}^m p_i(x, u) \delta_{x'_i(x, u)}(x'),$$

where $\delta_{x'_i(x, u)}$ is a Dirac centered at $x'_i(x, u)$ and $\sum_{i=1}^m p_i(x, u) = 1$.

Question 8.4 Provide such a finite-support kernel that approximates the transition kernel of the system (no need to do something complicated).

Hint: Use for instance a Monte-Carlo (aka. sample-based) approximation.

Question 8.5 Apply the LSPE algorithm to estimate the Q-function of the closed-loop system with policy π_{cl} . Use Question 8.4 to approximate the transition kernel of the system. To generate your data, use an exploration policy π_{exp} , e.g., use $\pi_{\text{exp}}(\cdot \mid x) \sim \mathcal{N}(K_{\text{cl}}x, \sigma_{\text{exp}}^2)$. Make also a reasonable choice of basis functions, motivated briefly. Comment your results.

Question 8.6 Consider the *LQR approximate model* obtained in Section 4, i.e., with objective (7). For this model, repeat Question 8.5 (but for the comparison, use the exact Q-function, which you can compute analytically).

8.2 Implementation of LSPE+PI

In this subsection, we use the parameters given in Table 1.

We can now combine the learning of the Q-function (based on LSPE above) with the policy improvement step, overall referred to as LSPE+PI

Question 8.7 For the setting of Question 8.5, implement the LSPE+PI algorithm, using π_{cl} as initial policy. Denote the resulting policy by π_{lspepi} . Repeat Question 3.2 for π_{lspepi} . Compare the averaged reward with that of the policies π_{cl} and π_{lqr} .

Question 8.8 For the setting of Question 8.6, implement the LSPE+PI algorithm, using π_{cl} as initial policy. Demonstrate the convergence of the gain sequence K_k to K_{lqr} (where K_{lqr} is the gain associated to π_{lqr}) by plotting the evolution of the error norm with k .

Finally, since the policies may violate the constraint $q_t \in [0, 1]$ for all $t \in \mathbb{N}$, we incorporate this constraint in the policy improvement step. Therefore, the policy improvement consists in solving the optimization problem

$$\pi_\theta(x) = \arg \max_u Q^\theta(x, u) \quad \text{s.t.} \quad 0 \leq [1 \ 0 \ 0]x + u \leq 1, \quad (9)$$

Question 8.9 Repeat Question 8.7 (i.e., using the true reward model of the system) but with the constrained policy improvement step discussed above.